

FEASIBILITY FOR SIDETCHANNEL DEVELOPMENT ON THE EAST SIDE OF REACH 1 OF THE JORDAN RIVER

FWCP SEED PROJECT 13.JOR.01

Prepared For:

Fish and Wildlife Compensation Program

Prepared By:

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The Pacheedaht First Nation

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EXECUTIVE SUMMARY

In 2013, D. Burt and Associates was awarded seed funding through the Fish and Wildlife Compensation Program (FWCP) to undertake a preliminary feasibility assessment of sidechannel development in the tidal reach of the Jordan River. The location examined was a bench on the East side of the River just downstream of the Jordan River Generating Station.

Field investigations indicated that the most promising path for sidechannel construction was a depression that runs the length of the bench adjacent to the valley wall. A level and rod survey along this depression indicated a potential sidechannel length of 430 m with an overall gradient of 1.33%. Sources of water for the sidechannel showed promise and included groundwater seepage from various locations along the bench sideslope, as well as 3 local tributaries. Of the 3 tributaries, the lower two (Creeks 1 and 2) would easily feed into the sidechannel, while the upper one (Creek 3) would need additional investigation to determine if local terrain permitted routing it to the sidechannel site. In terms of potential copper issues, Creeks 1 and 2 were clean of copper, while Creek 3 picked up copper from the vicinity of the old mine portal access road. Thus, if Creek 3 were to service the sidechannel, it would need to be rerouted from upstream of the portal access road.

The combination of good machine access, and promising water inputs from groundwater and local tributaries, suggested this site warrants movement to the next stage of feasibility assessment. This would include test pits to ensure no underlying bedrock, installation of perforated pipes to monitor groundwater input rates, testing of groundwater water quality, engineering assessment into the viability of bringing in mainstem flows or whether to design the channel as a purely groundwater and tributary fed installation. If these proved favourable, then a full topographic survey of the site and engineering drawings would follow.

At present, salmon production from anadromous reaches of the Jordan River is severely limited by a shortage of both spawning and rearing habitat. Development of a sidechannel at this site has the potential of creating an additional 430 m of new habitat that could support both spawning and rearing life stages. Furthermore, a sidechannel offers a protected and productive off-channel environment, which at present, is completely absent in the anadromous portion of Jordan River. Thus, development of this option offers substantial benefit to overall fish production from the Jordan system. Species anticipated to benefit most are coho salmon, however, the sidechannel may also provide spawning opportunities for pink and chum salmon, as well as spawning and rearing for anadromous cutthroat trout.

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1. INTRODUCTION

Historically the Jordan River supported an abundance of anadromous salmon and trout runs with a species mix that included coho, chum, and pink salmon, and steelhead and searun cutthroat trout. Coho and Chum runs declined suddenly in the early 1950's and became extinct by the late 1950's – early 60's. Pink salmon were last observed in 1970, after which they too became extinct (Burt 2012 (draft)). Six years of monitoring by the Water Use Plan studies provided insight into the demise of these runs (summarised in Burt 2014). Foremost among factors, was that most of the anadromous portion of the river was contaminated with copper leaching in from the top end of the accessible section. While copper levels did not deter successful spawning and incubation, they completely prevented occurrence of the more sensitive freshwater rearing stage. Thus species such as coho and steelhead have been unable to complete their life history and produce outgoing smolts. It seems likely that copper contamination was the main reason for the abrupt decline in salmon runs in the 1950's and 60's. The stragglers that hung on for some time, as well as the pink runs, were likely sustained by the original tailrace channel, which was a 500 m long outflow channel for the original generating station located on the east side of the river. This 500 m long channel provided a section of aquatic habitat free of copper contamination. However, when BC Hydro moved the plant to the west side of the river in 1971, and then decommissioned the channel, this last freshwater refuge was lost thus sealing the fate of remaining salmon runs.

This grim situation was dramatically improved beginning in January 2008 when BC Hydro initiated fish flow releases from Elliott Dam. The terms of the Jordan WUP called for a release of 0.25 m³/s, however, issues with the control mechanism necessitated locking the valve in the full open position, and this has resulted in release flows of roughly 0.30 – 0.40 m³/s depending on water elevation in Elliott Headpond. For the anadromous portion of the Jordan River, these releases have provided sufficient dilution of copper that rearing fish have colonized this previously dead zone. Furthermore, sampling of juvenile trout found that fish condition factor (weight to length ratio) was comparable to fish sampled upstream of the copper impacted zone (Burt 2013). This turn around has opened the possibility for undertaking habitat restoration and related projects that help to restore salmon and trout runs back to the Jordan River.

The purpose of this seed fund project was to investigate the feasibility of one restoration potential: that of constructing a sidechannel on a bench located in Reach 1 on the east side of the river. A sidechannel option has a number of attractions: 1) through use of tributary and groundwater inputs, it can provide aquatic environment free of copper contamination, 2) sidechannels are readily used by historically pertinent species such as coho, chum, and pink salmon, 3) sidechannels can be protected from mainstem flood events, and 4) production from sidechannels can be quite high resulting in a favourable cost-benefit ratio.

The scope of this project was to provide a preliminary feasibility assessment for developing a sidechannel on the east bench in Reach 1 of the Jordan River. This assessment was of a reconnaissance

level; test pits for water sources and engineering drawings were outside the scope, and would be the focus of future FWCP submissions if the current assessment provided positive results.

2. STUDY AREA

The anadromous part of the Jordan River contains 3 distinct reaches (Figure 1). Reach 1 is tidal and extends from the mouth to just upstream of the current tailrace outlet, a distance of 970 m. This reach is influenced by flows from the generating station, which exhibit wide fluctuations due to operation of the Jordan facility as a peaking power plant. The resultant flows appear to have scoured most gravels from this reach severely degrading its spawning capabilities. Reach 2 extends from the top of Reach 1 upstream for 330 m to a partial barrier falls. The copper contamination mentioned in the Introduction emanates from an exposed section of the east bank at the top of this reach. Reach 3 is 325 m in length and terminates at a 4.6 m rock falls, considered to be a complete barrier to anadromous migration. This third reach is steeper gradient, filled with extremely large boulders (4 m +), and very difficult for fish to ascend.

The area of interest for sidechannel development is a bench on the east side of the river in Reach 1 (Figure 1). Potential length following the foot of the valley wall is about 460 m, though meanders in construction could increase the overall length. Potential tributary water sources include unnamed Creeks 1, 2, and 3. Potential access is via an existing road which traverses the side of the valley wall and then swings north onto the bench.

3. METHODS

Assessment of the study area was undertaken on August 15, 21, and 22, 2014 by Dave Burt (D. Burt and Associates), and Helen and Jeff Jones (Pacheedaht First Nation fisheries technicians). Activities undertaken during these field trips included:

- Assessment of vehicle and machine access to the potential sidechannel development area.
- Reconnaissance of nearby creeks as potential water sources for the sidechannel, and examination of the base of the valley wall for groundwater sources.
- Collection of water samples from these creeks for lab analysis to detect any copper contamination (analysed for total and dissolved copper).
- Completion of a simple topographic survey of the likely path of the sidechannel using a builder's level and rod, tape measure, and laser rangefinder, and handheld GPS (Garmin Oregon).
- Installation of 3 benchmark pins along the traverse for future survey tie-in reference points.
- Collection of photographs along the traversed path, at the entrance/exit points, and of potential creek water sources.

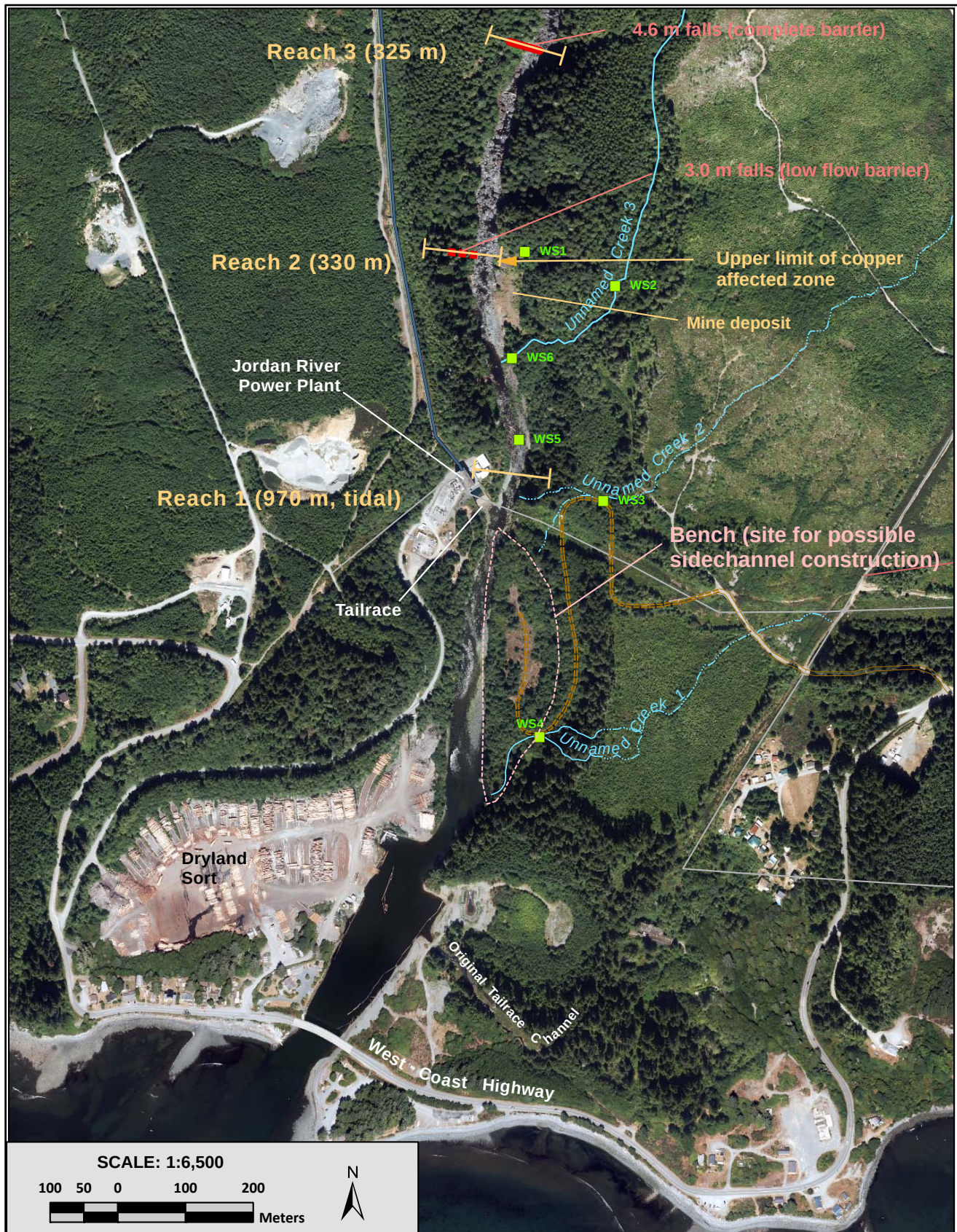


Figure 1. Map of the 3 anadromous reaches of the Jordan River. Also shown are the bench investigated for sidechannel development, the access road (brown parallel dashed lines), and locations of water sample sites (small green squares) (May 2013 orthophoto courtesy of CRD's WMS server).

Other activities included providing a PowerPoint presentation to the Pacheedaht Chief and Council and Queesto Community Forest (current owners of logging and dryland sort operations on the Jordan River) in order to provide a summary of the current status of the Jordan River and to garner support for this and future projects on the river. As well, other land owners were contacted to discuss the project including CRD (Stephen Henderson, 250-360-3176) and Western Forest Products (Brian Marcus, 250-720-4226)

Lastly, a tour of the lower river and sidechannel option was provided on October 23, 2013 to representatives of DFO (Margaret Wright), the Comptroller (Gypsy Fisher), and BC Hydro (Ian Dodd).

4. RESULTS

Access

Machine access to the potential sidechannel site is by an existing gravel road on the east side of the river (Figure 1). This road is in good shape and the margins recently machine brushed to the point where it passes under the BC Hydro power lines for the second time (just past unnamed creek 2). For the remaining 400 m, the road descends down the sideslope of the bench wall, and the margins are growing in with shrubs and alder shoots, and will need to be brushed. There is also the remains of a fallen tree that will need to be removed by chainsaw. Lastly, where the road ascends onto the bench (at the switchback), there is a plugged culvert and flows that occur along the depression at the base of the sideslope have partially eroded away the access road (Photo 6 in Appendix I). Once on the bench the ground is flat, stable, and open, giving good machine access to most locations on the bench.

Longitudinal Profile

A longitudinal traverse of the sidechannel site was completed on August 20 – 21, 2013. The traverse started in the Jordan mainstem adjacent to the top of the site, progressed up the bank to where the sidechannel would start, and then followed what appeared to be the best path for sidechannel development. This was essentially a muddy hollow that runs the full length of the bench adjacent to the valley wall. After crossing the access road, the traverse jogged south by 48 m to connect with the channel of unnamed Creek 1. From this point it followed the creek channel for 90 m to its mouth with the Jordan River. A plan view of the traverse, and the resultant longitudinal profile are shown in Figure 2. Pertinent statistics generated from the traverse data are shown in Table 1.

The total length of the traverse from beginning to end was 460 m, however, the length of the potential sidechannel portion was 430 m. The overall gradient for the sidechannel portion was 1.33%. The height of ground where a mainstem intake might occur is 4.674 m above the Jordan River channel bed. The high water mark at this location is 0.8 m above the stream bed. Photos of various locations along the traverse are given in Appendix I.

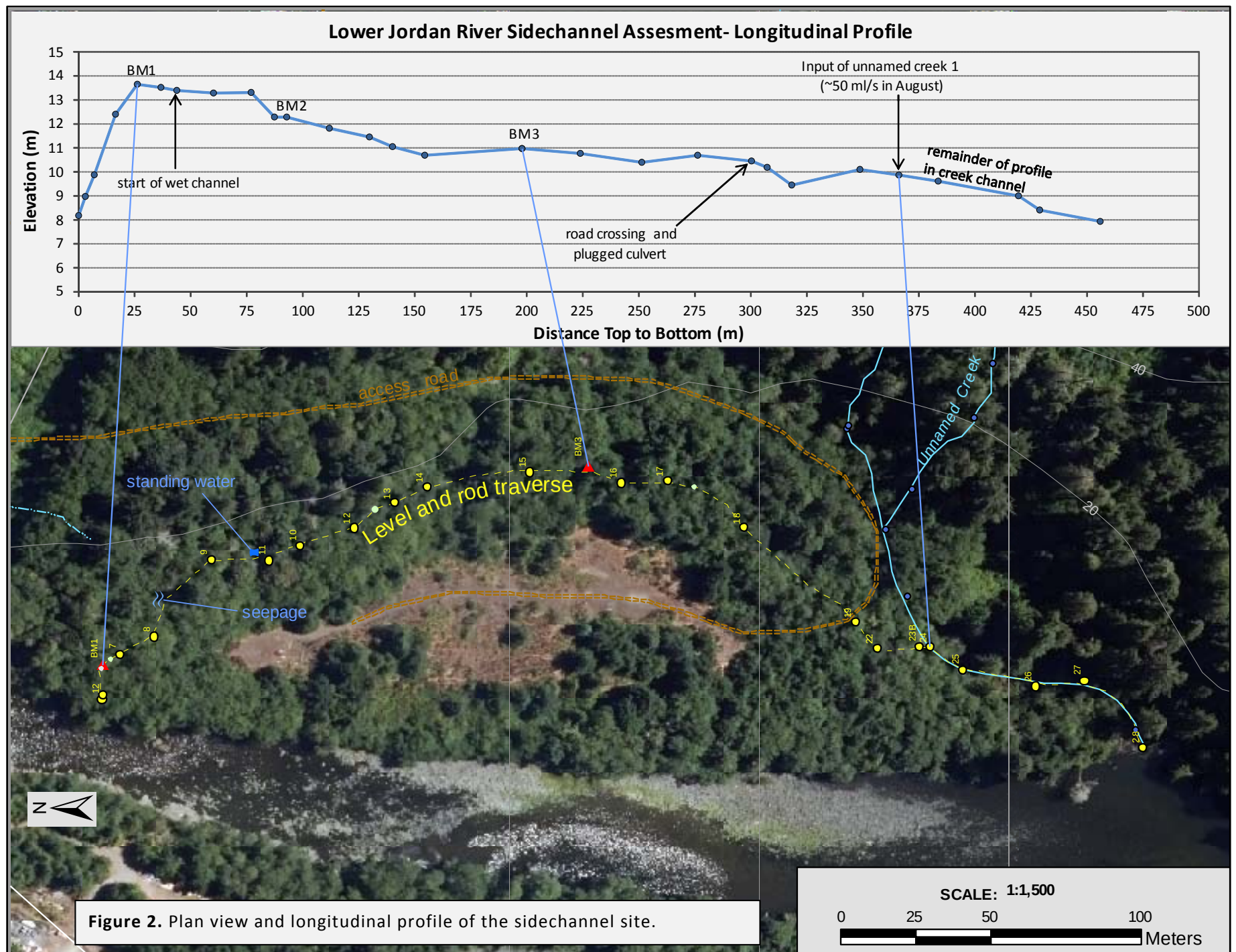


Figure 2. Plan view and longitudinal profile of the sidechannel site.

Table 1. Summary of the level and rod traverse results.

Length of proposed channel based on traverse path	430 m
Overall gradient of proposed channel	1.33%
Height of land above the river bed at the top end	5.476 m
Height of the high water mark above the stream bed	0.802 m
Height of land above high water mark at the top end	4.674 m

Water Sources and Water Quality

Since the field investigation was conducted in August, it probably captured the study area when groundwater and stream flows were lowest. At this time the traversed area was damp or muddy throughout with the occasional puddle of standing water (temperature 11.9 °C on August 15, 2013). Groundwater seepage from the sideslope was evident at multiple locations along the traverse, and emanated as surface flows at some locations (temperature 12.1–13.7 °C, pH 7.0, August 15, 2013). Thus, groundwater sources look promising, however test pits will be need to be part of future works in order to measure the rate of inflows.

It is noteworthy, that during a site visit on October 23, 2013, the surveyed route was completely under water along its length, with flowing water wherever there was gradient. Thus, there appears to be sufficient water to fill a sidechannel during wetter times of the year (Appendix I, Photo 8).

In addition to groundwater, there are 3 small creeks which may be used to supply flows to a future channel. These have been designated as unnamed Creeks 1, 2 and 3 in Figure 1. On August 15, 2013, the lowermost of these (Creek 1) had a temperature of 12.7 °C, pH of 6.8, and estimated discharge varied from 50–100 ml/s, depending substrate porosity. Bank scour observed on this creek indicated that flows increase substantially during rainfall events. The lower end of this creek has a beautiful setting with large mature timber on the sideslope, short sections of potential spawning gravels, and an outlet that tapers gently into the tidal section of the Jordan River. Photos of this creek are provided in Appendix I (Photo 9, Photo 10, and Photo 11).

Creek 2 originates in the logged area to the east of the Jordan River. It enters a treed area northeast of the bench access road, passes over waterfall, and then parallels the access road before diverging in two directions (Figure 1). In the logged area, riparian vegetation was removed and water retention diminished resulting in a stream that is flashy and heavily eroded (the mine portal access road crossing is completely washed out and impassable by vehicle). Where the creek parallels the bench access road, erosion and deposition of gravels from past freshets are evident, however, during the August surveys, surface flows were only about 2 ml/s. This creek used to enter the Jordan mainstem via a vertical embankment opposite the powerhouse, however some of the flows now appear to flow south toward the top of the sidechannel though no defined channel was found. A logical course of action would be

to collect all flows from this creek, putting them in a defined channel that empties into the top end of the sidechannel. Photo 13 in Appendix I shows this creek at the falls adjacent to the bench access road.

Creek 3 is the furthest north of the creeks examined (Figure 1), and in all the years of work on the Jordan by the author, has always had a consistent surface flow of about 100 ml/s in the driest part of summer. This creek originates near the pit and flows in a southerly direction within a deeply incised gully (possibly a result of scour by the mine washout in 1964). Where it crosses the mine portal access road it has eroded half of the road away exposing what looks to be piping that once carried tailings from the mine to the ocean outfall site. Interestingly, water samples from above the road crossing showed no copper contamination, while those from below showed excessive copper levels. Thus the creek is picking up copper from around the road crossing. If this creek is to be rerouted to service the sidechannel, it needs to be taken off from upstream of the road crossing. This may also benefit the Jordan mainstem by slowing the rate of copper input.

Due to copper contamination problems on the lower Jordan River, water samples were collected from various sites and analyzed (North Island Labs) for dissolved and total copper (detection limit 0.001 mg/L). It was important to determine whether potential water sources for the sidechannel had copper contamination issues. Seven samples were collected from 5 sites during field trips spanning August 15 – November 5, 2013. Locations of the 5 sample sites are shown in Figure 1. These included one site each on Creeks 1 and 2, two sites on Creek 3 (above and below the mine access road), one site capturing outflows from a pipe protruding from the cemented mine portal, and one site in the Jordan mainstem downstream of Creek 3. Results from copper analysis are shown in Table 2.

Table 2. Summary of copper analysis results for water samples collected during August 15 to November 5, 2013 field investigations.

Location	Date	Name	Dissolved Cu (mg/L)	Total Cu (mg/L)
Creek 1 beside the sidechannel access road	Aug. 22, 2013	WS4	< 0.001	0.001
Creek 2 where it comes in adjacent to sidechannel access road	Aug. 22, 2013	WS3	< 0.001	0.001
Creek 3 upstream of the mine access road (at the falls)	Aug. 15, 2013	WS2	0.002	0.001
Creek 3 downstream of the mine access road (40 m upstream of its mouth)	Sept. 9, 2013	WS6	0.012	0.013
Seepage from the mine portal pipe during the summer dry period	Aug. 15, 2013	WS1	0.285	0.375
Runoff from the mine portal pipe after rain	Nov. 5, 2013	WS7	0.419	0.490
Jordan mainstem downstream of Creek 3 (WQ1 site in Burt 2012)	Sept. 9, 2013	WS5	0.006	0.008

Creeks 1 and 2 showed no sign of copper contamination, while as previously mentioned, Creek 3 was clean above the portal access road, but contaminated below this road. Samples WS1 and WS7 indicated that copper levels from water emanating from the portal pipe increase during rainfall events (dissolved copper increased from 0.285 mg/L to 0.419 mg/L). Sample WS5 from the Jordan mainstem was comparable to previous readings from this site since initiation of fish flows from Elliott Dam in 2008 (see site WQ1 in Table 2 of Burt 2012). These results indicate that Creeks 1 and 2, and Creek 3 above the portal road, are viable sources of water input for the sidechannel in terms of copper concerns.

Sources of Material

Potential materials required for sidechannel construction include riprap, gravel, and coniferous large woody debris (LWD) and root wads. There is a quarry off the West Coast Highway northwest of the generating station that may be able to supply gravel (visible in Figure 1) and BC Hydro has a stockpile of riprap in the same area. In addition, Queesto Community Forest have expressed an interest in assisting Jordan River restoration works through supply of in-kind machine support as well as materials where possible (e.g., riprap, gravels, and LWD).

5. DISCUSSION

The results of this seed project provide preliminary affirmation for development of a sidechannel on the Reach 1 bench, and that the project warrants moving to the next step of a more detailed engineering feasibility assessment.

One uncertainty the engineering assessment would need to resolve is whether the sidechannel would be able to receive water from the mainstem, or would be a purely groundwater and creek fed sidechannel. A possible problem with bringing in mainstem water is that the height of land at the top of the site may be too high for a direct mainstem link. It might be possible to alleviate this problem by bringing in mainstem water from an intake located upstream and carried via pipe to the depth of the cut at the upstream end of the channel. Another possibility would be to bring in water via a pipe connected to the penstock. The penstock is already equipped with a valve from when BC Hydro supplied water to the village on the east side of the river. This option would require a high pressure valve at the connection point, as well as support from BC Hydro.

If mainstem inputs are not possible, then the sidechannel would need to be built based on flows supplied by groundwater and the local tributaries described in this report. It is possible that this is the better option in that it allows the channel to be built with a higher overall gradient, and it offers an aquatic environment free of any copper issues. Observations to date suggest that the groundwater supply to the site is promising, but test pipe monitoring is needed to confirm this observation. Creeks 1 and 2 are also promising candidates for augmenting sidechannel flows. Creek 3 may also be possible as a water source, but a topographic survey is needed for the terrain between it and the sidechannel.

Potential benefits of developing a sidechannel on the lower Jordan River are substantial. Currently, most spawning and rearing by anadromous species returning to the Jordan River is limited to the 330 m of Reach 2. Furthermore, this reach was estimated to contain only 144 m² of spawning habitat (Cascadia Biological Services 2013). Thus anadromous production from the Jordan River is severely limited by a shortage of both spawning and rearing habitat. Development of this sidechannel option has the potential of creating 430 m of new habitat with both spawning and rearing capabilities. Moreover, it has the potential of providing protected off-channel habitat that can be very productive, and is currently non-existent within the anadromous portion of the Jordan River. Even if it is a groundwater and tributary fed sidechannel, there is likely to be sufficient flow during fall rains to support spawning by coho and chum salmon in higher gradient sections. Additionally, the proposed outlet (Creek 1) is low gradient, within a protected alcove, and already contains numerous pieces of large woody debris (see Photo 11). These features offer an attractant for any coho fry rearing in the tidal section of the mainstem, and the fry could potentially move into lower end of the channel to take advantage of high quality rearing habitat, while avoiding potential flushing flows from turbine operation. In terms of species use, coho are likely to benefit most, however, the sidechannel may also provide spawning habitat for pink and chum salmon, as well as spawning and rearing for anadromous cutthroat trout (once reported to be abundant in the lower Jordan River).

The sidechannel would also provide a productive environment for aquatic invertebrates, refugia for amphibians, and substrates for riparian and aquatic indigenous plant species. Lastly, proximity to the West Coastal Highway, ease of access, and setting of the bench, are factors conducive to development of the site for cultural, educational, and recreational activities.

Though copper issues in the mainstem have been dramatically improved since BC Hydro initiated flow releases from Elliott Dam, it is nevertheless preferable to remove the sources of copper contamination. To this end, plans are underway to form a Jordan River user group round table this summer (2014), and Teck Resources (formerly Cominco) has agreed to come to the table to discuss remediation of the copper issues. It is also envisioned that this round table would serve as a platform to share information, disseminate knowledge, and be the guiding mechanism for restoration works on the river. It is encouraging that the Pacheedaht First Nation are committed to recovery of the Jordan River, and that Queesto logging has already expressed an interest in assisting with future restoration initiatives.

6. RECOMMENDATIONS

Given results of this seed project, it is recommended that the sidechannel option move to the next stage, and receive a more comprehensive engineering feasibility assessment. This should include test pits to assess underlying material, installation of groundwater pipes to measure groundwater input rates, lab analysis on the quality of groundwater, a detailed topographic survey (total station and/or

survey grade GPS), assessment of options for mainstem inputs, and survey of rerouting paths for Creeks 1, 2 and 3.

7. REFERENCES

Burt, D.W. 2012 (draft). Summary of water quality and biological data for anadromous reaches of the Jordan River. Prepared for BC Hydro, Fish and Wildlife Compensation Program, John Hart Generating Station, Campbell river, BC. 51 p.

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Cascadia Biological Services. 2013. Jordan River spawning assessment and enumeration, Year 6 (Sept 2010 - May 2011). WUP Monitor JORMON#3. BC Hydro, Water Use Planning, Burnaby, BC. 54 p.

8. ACKNOWLEDGEMENTS

The author is especially grateful for the ongoing assistance provided by Helen and Jeff Jones of the Pacheedaht First Nation fisheries section. Many thanks for your assistance in the field, for undertaking the fall 2014 snorkel survey, and for untiring support of all things Di Di Daht (Jordan River). Thanks also to the staff and Jordan River Generating Station (Dwayne Walsh, Lee Hamilton, and Al Hall) for providing lockout and for ongoing anecdotal observations of fish activities.

This project was made possible through the financial support of the Fish and Wildlife Compensation Program, administered on behalf of program partners BC Hydro, the Province of BC, and Fisheries and Oceans Canada.

APPENDICES

Appendix I. Selected Photos.

Appendix II. Financial Statement (Statement of income and expenditures-form attached

Appendix IV. Confirmation of FWCP Recognition – Presentation to Pacheedaht Chief and Council

Appendix I. Selected Photos.



Photo 1. View of the left bank of the Jordan River at the upstream end of the bench opposite the tailrace. The top of the green algae layer marks the height of water at high tide, while the rock/moss boundary above this marks the height of water during turbine flows (August 21, 2013).



Photo 2. View of the Jordan mainstem embankment at the start of the traverse. The water's edge is at the base of the photo (August 21, 2013).



Photo 3. View of muddy area with standing water near the top end of the hollow (near position P08 on the traverse) (August 21, 2013).



Photo 4. View of the muddy depression between traverse points 15 and 17 (August 21, 2013).



Photo 5. View of the depression at benchmark 3 (BM3 is a spike in the base of the large tree stump below the orange marker paint) (August 21, 2013).



Photo 6. Plugged culvert and washout where the access road crosses the depression (August 21, 2013).



Photo 7. Groundwater seepage emanating from the sideslope at the top end of the site (August 15, 2013).



Photo 8. View of the depression on October 23, 2013. Groundwater inputs inundated the depression at this time of year.



Photo 9. Point where the traverse intercepted Creek 1 (August 21, 2013). The direction of flow is away from the camera.

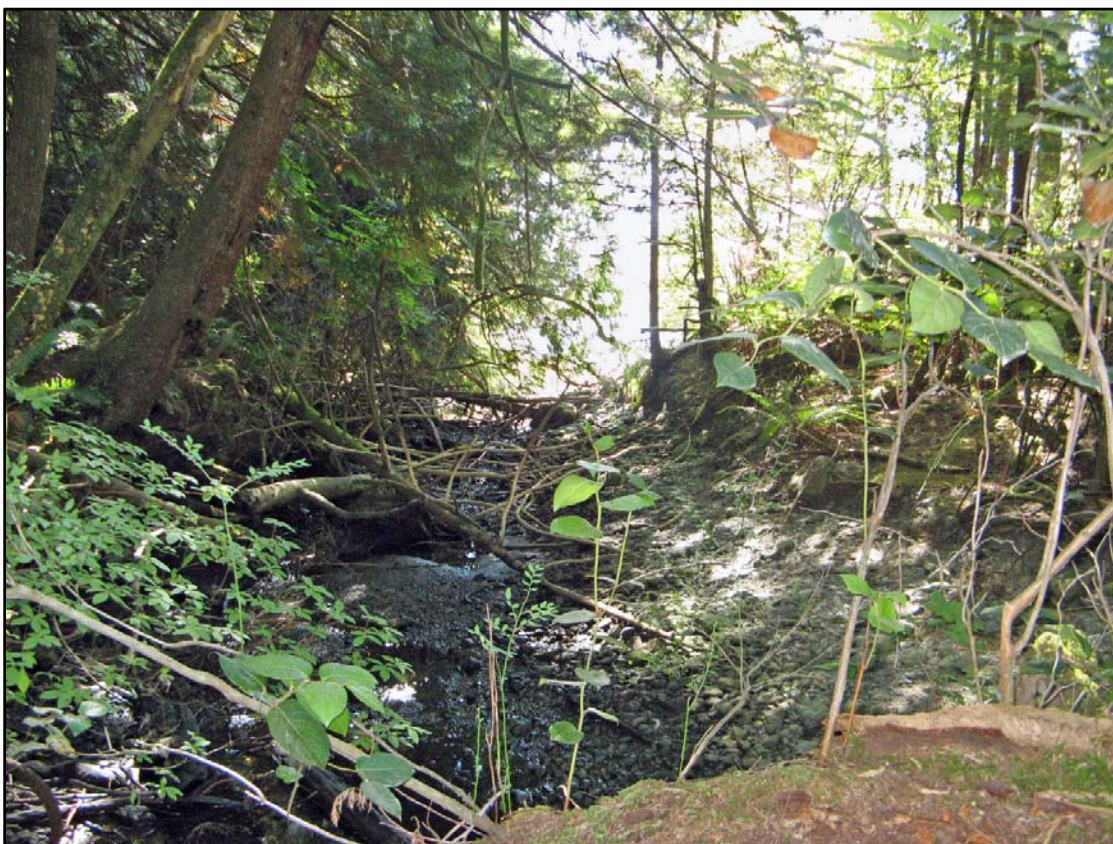


Photo 10. View of Creek 1 looking toward its confluence with the Jordan River (background) (August 21, 2013).



Photo 11. View of the mouth of Creek 1. The creek enters at a protected alcove in the mainstem where woody debris has accumulated (August 22, 2013). Such sites are a natural attractant for any coho fry rearing in the tidal reach.



Photo 12. Creek 3 at the waterfall located 120 m upstream of the mine portal access road (August 15, 2013).



Photo 13. Creek 2 at the falls located adjacent to the bench access road. The creek was just a trickle on the survey date but there was erosion and gravel deposition where it parallels the access road (August 22, 2013).



FISH AND WILDLIFE COMPENSATION PROGRAM

Financial Statement Form

Project # _____

	BUDGET		ACTUAL	
	FWCP	Other	FWCP	Other
INCOME				
<i>Total Income by Source</i>				
Grand Total Income (FWCP + other)				
	BUDGET		ACTUAL	
	FWCP	Other	FWCP	Other
EXPENSES				
Consultant Fees				
(List others as required)				
MATERIALS & EQUIPMENT				
Equipment Rental				
Materials Purchased				
Travel Expenses				
Permits				
(List others as required)				



FISH AND WILDLIFE COMPENSATION PROGRAM

Financial Statement Form – Continued

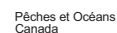
ADMINISTRATION				
Office Supplies				
Photocopies & printing				
Postage				
(List others as required)				
Total Expenses				
Grand Total Expenses (FWCP + other)				
BALANCE (Grand Total Income – Grand Total Expenses)				
	<i>The budget balance should equal \$0</i>		<i>The actual balance might not equal \$0</i>	

* Any unspent FWCP financial contributions are to be returned to:

Fish and Wildlife Compensation Program
c/o BC Hydro
11th floor
6911 Southpoint Drive
Burnaby, BC
V3N 4X8
Attention: Lorraine Ens

For more information visit fwcp.ca.

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Performance Measures

Using the performance measures applicable to your project, please indicate the amount of habitat anticipated to be restored/enhanced for each of the specified areas (e.g. riparian, tributary, mainstream). The same table will be used in the final report to summarize project results.

Performance Measures – Target Outcomes													
Project Type	Primary habitat benefit targeted of project (sq.m.)	Primary Target Species	Estuarine	In-stream Habitat – Mainstream	In-stream Habitat – Tributary	Riparian	Reservoir Shoreline Complexes	Riverine	Lowland Deciduous	Lowland Coniferous	Upland	Wetland	Other
Impact Mitigation													
Fish passage technologies	Area of habitat made available to target species		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Drawdown zone revegetation/ stabilization	Area turned into productive habitat		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Wildlife migration improvement	Area of habitat made available to target species		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Prevention of drowning of nests, nestlings	Area of wetland habitat created outside expected flood level (1:10 year)		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Habitat Conservation													
Habitat conserved – general	Functional habitat conserved/replaced through acquisition and management		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Habitat conserved - general	Functional habitat conserved by other measures (e.g. riprapping)		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Designated rare/special habitat (subset)	Rare/special habitat protected		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Performance Measures – Target Outcomes - Continued

Project Type	Primary habitat benefit targeted of project (sq.m.)	Primary Target Species	Estuarine	In-stream Habitat – Mainstream	In-stream Habitat – Tributary	Riparian	Reservoir Shoreline Complexes	Riverine	Lowland Deciduous	Lowland Coniferous	Upland	Wetland	Other
Maintain or Restore Habitat forming process													
Artificial gravel recruitment	Area of stream habitat improved by gravel placement		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Artificial wood debris recruitment	Area of stream habitat improved by LWD placement		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Small-scale complexing in existing habitats	Area increase in functional habitat through complexing		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Prescribed burns or other upland habitat enhancement for wildlife	Functional area of habitat improved		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Habitat Development													
New habitat created	Functional area created		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Other													
			☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
			☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
			☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
			☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Appendix IV. Confirmation of FWCP Recognition

Presentation to Pacheedaht Chief and Council

[to be sent separately due to file size]